

Exercise 3

Stability

3.1 Modelling and control of a motion system

Again consider the classic mass-spring-damper-mass system from Exercise 1.5, depicted in figure 1.1. For this exercise we consider the motor feedback case, i.e. the input is F and the output is x_1 , using the parameters mentioned in Exercise 1.5.

- Make a simple stabilizing controller for this system such that a bandwidth (0db of open-loop) of 20 Hz is achieved, and $|S| < 6\text{dB}$ (and reasonable phase and gain margins).
- Implement the system and the controller in a Simulink (closed loop) block scheme.
Hint: you can use blocks like ‘Transfer Fcn’ (in ‘Simulink - Continuous’) or ‘LTI system’ (in ‘Control System Toolbox’)
- Use a step as a reference signal, and perform a simulation. Plot the resulting error.
- Try to find out what the influence of the frequency domain specifications *bandwidth* and *phase margin* is on time domain performance like overshoot, rise time and settling time. Do this by designing and applying different controllers (with varying bandwidth and phase margins) and comparing the resulting error plots. Try to explain these influences.
- Apply the controller from the first step to the non co-located plant $H_2(s)$. Is the resulting closed loop stable? Explain why or why not.

3.2 Inverted pendulum

Consider the inverted pendulum problem, see figure 3.1. The objective of this problem is to keep the pendulum in upward vertical position $\phi = 0$, by only applying a horizontal force F to the cart. It is easy to see that this system is non-linear. However, around the equilibrium $\phi = 0$ the system can be linearized, resulting in the following transfer function (you might want to show this yourself):

$$H(s) = \frac{\Phi(s)}{F(s)} = \frac{s}{-\frac{1}{6}(4M + m)Ls^3 - \frac{2}{3}Lbs^2 + g(M + m)s + gb}, \quad (3.1)$$

where L is the length of the pendulum and b is the friction between the cart and the floor. Furthermore, assume that

$$M = 0.05, \quad m = 0.04, \quad b = 0.02, \quad g = 9.81, \quad L = 0.15.$$

Now answer the following questions:

- Draw a Bode diagram of this system. By looking at this plot, argue whether this system is stable or unstable. Verify your result by calculating the poles of (3.1).
- Consider the simple controllers $C_1 = 1$ and $C_2 = -0.55$. Draw the Nyquist plots of the open loops C_1H and C_2H . Would you label the closed loops stable or unstable? Why? Check your answer by making time plots of the step response of the closed loop $T_i = \frac{C_iH}{1+C_iH}$. *Hint: read Section 6.3 (Nyquist Stability Criterion) of the book of Franklin and Powell.*
- Design a stabilizing controller with a bandwidth of 10 Hz. Prove the closed loop stability by making a step response plot and analyzing the Nyquist plot properly. *Hint: use Franklin's formula $Z = N + P$, and design your controller such that you have the right amount of counterclockwise encirclements of the point -1.*
- Suppose the cart is on a slope with a small (unknown but constant) angle, causing the cart to drift away. Show how this changes the closed loop by drawing a block scheme. Does this change the closed loop stability? Is your controller still able to keep the pendulum in upright position?

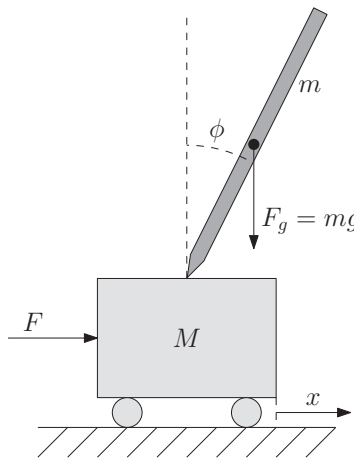


Figure 3.1: Schematic of the inverted pendulum example

3.3 Non-minimum phase systems

Consider the following third order systems

$$H_1(s) = 1000 \frac{s + 6}{s^3 + 20s^2 + 5000s} \quad \text{and} \quad H_2(s) = 1000 \frac{s - 6}{s^3 + 20s^2 + 5000s}.$$

- Draw the frequency response of both systems in one Bode diagram.
- Compare the magnitude and phase diagrams. What difference do you notice? Explain what causes this difference.
- System H_2 is called *non-minimum phase*. Explain why.
- Design a simple stabilizing controller C_1 for H_1 . Put the bandwidth around 30Hz and attain a robustness margin $|S| < 3.5\text{dB}$. What's your phase margin?
- Apply the same controller C_1 to H_2 and draw the open loop Bode diagram. What's the phase margin?
- Draw the Nyquist curve of $C_1(j\omega)H_2(j\omega)$. Would you label this system stable or unstable? Why? Verify your result by drawing a time response plot.
- Design a new stabilizing controller C_2 for H_2 . Try to achieve a high as possible bandwidth while attaining $|S| < 6\text{dB}$. What is the bandwidth-limiting factor in this case?